

Certified Exclusion of Order-Twelve, Parameter-Three Counterexamples to the γ - θ Conjecture

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Abstract

In the one-guard-moves eternal domination game, an attack is made at an unoccupied vertex and exactly one guard moves along one edge to the attacked vertex. The γ - θ conjecture asserts that $\gamma(G) = \gamma^\infty(G)$ implies $\gamma(G) = \theta(G)$. We establish the following certificate-backed finite result: no finite simple graph G on 12 vertices satisfies

$$\gamma(G) = \gamma^\infty(G) = 3 < \theta(G).$$

The result includes disconnected graphs. The mathematical reduction sends any putative graph to one of three overlapping cases in which $\overline{G}$ contains a hub-free induced C_5 , C_7 , or C_9 . Each case is represented by a byte-bound propositional formula encoding the graph, a nonempty closed family of dominating triples, and valid coloring clauses implied by the obstruction to a three-clique partition. Addition-only reverse-unit-propagation refutations of all three formulas were replayed independently. We give the coverage proof, exact formula and proof hashes, and reproduction commands. This does not exclude parameter $k \geq 4$ at order 12, does not exclude larger-order counterexamples, and does not resolve the universal conjecture.

1 Introduction

For a finite simple graph G , let $\gamma(G)$ be its domination number and let $\theta(G)$ be the minimum number of cliques in a partition of $V(G)$. In the standard one-guard eternal domination game, guards occupy a dominating set, an attacker selects an unoccupied vertex, and the defender moves one adjacent guard to the attacked vertex while retaining a dominating configuration. The least number of guards permitting defense against every finite adaptive attack sequence is denoted by $\gamma^\infty(G)$.

Klostermeyer and MacGillivray stated the implication $\gamma = \gamma^\infty \Rightarrow \gamma = \theta$, with a proof that was later found to be incomplete [3, 4]. Klostermeyer and Mynhardt consequently posed the existence of a graph with

$$\gamma(G) = \gamma^\infty(G) < \theta(G) \tag{1.1}$$

as an open problem [4, 5]. This is now commonly called the γ - θ conjecture. MacGillivray, Mynhardt, and Virgile reported an exhaustive search finding no counterexample through order 11 [6]. Several proper graph classes are also known, but those results do not settle (1.1) universally; see, for example, the recent planar result [7].

Our main result is one precisely delimited slice at the next order.

Theorem 1.1. *There is no finite simple graph G with $|V(G)| = 12$ such that*

$$\gamma(G) = \gamma^\infty(G) = 3 < \theta(G).$$

Theorem 1.1 is not the assertion that there is no counterexample of order 12: the common value in (1.1) could be at least four. It is also not an extension of the published all-parameter enumeration through order 11. Its proof combines graph-theoretic reductions with exact finite certificates. Section 2 gives the reductions, Section 3 proves that three complement templates are exhaustive, Section 4 gives the graph-to-formula coverage argument, and Section 5 records the checked refutations. Appendix A gives the reproducibility record.

2 Definitions and elementary reductions

All graphs are finite, simple, and undirected. For $v \in V(G)$, let $N(v)$ and $N[v]$ denote its open and closed neighborhoods. A set $D \subseteq V(G)$ dominates G when $\bigcup_{v \in D} N[v] = V(G)$. Let $\alpha(G)$ be the independence number, and let $i(G)$ be the minimum cardinality of a maximal independent set. The identity

$$\theta(G) = \chi(\overline{G}) \quad (2.1)$$

will be used throughout.

Definition 2.1. An *eternal dominating family of size k* is a nonempty family $\mathcal{D} \subseteq \binom{V(G)}{k}$ such that every $D \in \mathcal{D}$ dominates G , and for every $D \in \mathcal{D}$ and every attacked vertex $r \notin D$, there is some $u \in D \cap N(r)$ for which

$$(D - \{u\}) \cup \{r\} \in \mathcal{D}. \quad (2.2)$$

The minimum such k is $\gamma^\infty(G)$.

Thus attacks at occupied vertices are not part of the model, and (2.2) moves exactly one guard along one edge. No all-guards-move parameter is used in this paper.

Lemma 2.2 (independent-set forcing). *Let $\mathcal{D}$ be an eternal dominating family of k -sets, and let I be independent. From every $D \in \mathcal{D}$, attacks at currently unoccupied vertices of I force a member $D' \in \mathcal{D}$ with*

$$|D' \cap I| = \min\{|I|, k\}.$$

In particular, if $|I| = k$, then $I \in \mathcal{D}$.

Proof. Attack a vertex $r \in I - D$. A responding guard cannot move from another vertex of I , since I is independent. The response therefore increases $|D \cap I|$ by one. Repeat until either all guards or all vertices of I are occupied. $\square$

Proposition 2.3 (parameter chain). *For every nonempty graph G ,*

$$\gamma(G) \leq i(G) \leq \alpha(G) \leq \gamma^\infty(G) \leq \theta(G). \quad (2.3)$$

Consequently, if $\gamma(G) = \gamma^\infty(G) = k$, then

$$\gamma(G) = i(G) = \alpha(G) = \gamma^\infty(G) = k. \quad (2.4)$$

Proof. Every maximal independent set dominates, and a maximum independent set is maximal, proving the first two inequalities. If an eternal family has $k < \alpha(G)$ guards, Lemma 2.2 places all guards in a maximum independent set while leaving another vertex of that set unoccupied; the next attack there has no adjacent guard. Hence $\alpha \leq \gamma^\infty$.

For a partition of $V(G)$ into t cliques, place one guard in each part. An attack is answered by moving the guard within the attacked vertex's clique, so $\gamma^\infty \leq t$. Minimizing gives the final inequality. Equality (2.4) follows by squeezing. $\square$

We need three further standard consequences. Proofs are included because they determine the exact search universe.

Lemma 2.4 (induced-subgraph monotonicity). *If J is a nonempty induced subgraph of G , then $\gamma^\infty(J) \leq \gamma^\infty(G)$.*

Proof. Let $\mathcal{D}$ be an eternal family in G , and among its configurations retain those maximizing the number m of guards in $V(J)$. Project these configurations to $V(J)$. For an attack inside J , a response moving a guard from outside J would increase the number of guards in $V(J)$, contradicting maximality. Hence every response stays inside J . Because J is induced, it uses an edge of J , preserves the maximum m , and projects to the required one-guard successor.

The same response argument shows that every projected state dominates J : each unoccupied vertex of J has the responding guard as a neighbor in the projected state. If $m = 0$, an attack at any vertex of the nonempty graph J would necessarily move a guard into J , again contradicting maximality. Thus $m \geq 1$, and the projected family is a nonempty eternal family of m -sets. $\square$

Lemma 2.5 (component additivity). *If $G_1, \dots, G_t$ are the nonempty components of G , then*

$$\gamma(G) = \sum_j \gamma(G_j), \quad \gamma^\infty(G) = \sum_j \gamma^\infty(G_j), \quad \theta(G) = \sum_j \theta(G_j). \quad (2.5)$$

Proof. The assertions for γ and θ follow by restriction to components and union of componentwise solutions. Products of eternal families give the upper bound for γ^∞ .

For the reverse bound, choose one guard-count vector $(|D \cap V(G_1)|, \dots, |D \cap V(G_t)|)$ occurring in an eternal family, and retain only configurations with that vector. A legal move cannot cross components, so this subfamily is closed under all attacks. Its projection to G_j is an eternal family with $|D \cap V(G_j)|$ guards. Summing the componentwise lower bounds proves equality. $\square$

Lemma 2.6. *For odd $n \geq 5$,*

$$\gamma^\infty(\overline{C_n}) = 3. \quad (2.6)$$

Proof. The upper bound follows from $\theta(\overline{C_n}) = \chi(C_n) = 3$. For the lower bound, label C_n cyclically by $0, \dots, n-1$. A pair fails to dominate $\overline{C_n}$ exactly when its two vertices have cyclic distance two. If an eternal two-family existed, Lemma 2.2 would put $\{0, 1\}$ in it. From $\{0, d\}$, for odd $1 \leq d \leq n-6$, attack $d+2$. Moving the guard at 0 leaves the nondominating pair $\{d, d+2\}$, so closure forces $\{0, d+2\}$. Iteration reaches $\{0, n-4\}$. An attack at $n-2$ leaves a pair at cyclic distance two whichever guard moves, a contradiction. For $n = 5$, this final attack applies to the initial pair. $\square$

This value is classical [2]; the argument above fixes the convention used here. We also need the following small cycle value.

Lemma 2.7. $\gamma^\infty(C_7) = 4$.

Proof. A partition into three edges and one singleton gives the upper bound four. Suppose an eternal family of dominating triples exists. A dominating triple has cyclic gap multiset $\{2, 1, 1\}$ or $\{2, 2, 0\}$. Every triple of the second type is equivalent to $\{0, 1, 4\}$; attacking 3 forces the move from 4, after which vertex 5 is undominated. Thus no second-type triple belongs to the family.

The maximum independent set $\{0, 2, 4\}$ belongs to the family by Lemma 2.2. Attack 1. Moving the guard at 0 leaves vertex 6 undominated; moving the guard at 2 produces the forbidden second-type triple $\{0, 1, 4\}$. This contradiction proves the lower bound. $\square$

Lemma 2.8 (minimum counterexample parameter). *Every graph satisfying (1.1) has common value*

$$\gamma(G) = \alpha(G) = \gamma^\infty(G) \geq 3.$$

Proof. Equality follows from Proposition 2.3. Common value one makes G complete. Suppose the common value is two and $\theta(G) > 2$. For $H = \overline{G}$, $\omega(H) = 2 < \chi(H)$. The Strong Perfect Graph Theorem [1] gives an odd hole or odd antihole in H . An odd hole in H gives an induced odd antihole in G , contradicting Lemmas 2.4 and 2.6. An odd antihole in H gives an odd hole in G . At length at least seven this contradicts $\alpha(G) = 2$, and at length five it again contradicts Lemma 2.6, because C_5 is self-complementary. $\square$

3 The exhaustive three-template split

Call a vertex outside an induced cycle a *hub* if it is adjacent to every vertex of the cycle. A cycle is *hub-free* if it has no hub.

Proposition 3.1. *If a graph G satisfies the hypothesis of Theorem 1.1, then G is connected.*

Proof. Let $G_1, \dots, G_t$ be its components and put $a_j = \gamma(G_j)$, $b_j = \gamma^\infty(G_j)$, and $c_j = \theta(G_j)$. By (2.3) and (2.5), $\sum a_j = \sum b_j = 3 < \sum c_j$. Nonnegativity of $b_j - a_j$ forces $a_j = b_j$ for every j , while the strict inequality gives a component G_q with $b_q < c_q$. This component is itself a counterexample, so Lemma 2.8 gives $a_q \geq 3$. Since all nonempty components have positive domination number and $\sum a_j = 3$, $G_q = G$. $\square$

Proposition 3.2 (odd-wheel obstruction). *If $\gamma^\infty(G) = 3$, then $\overline{G}$ contains no induced odd wheel.*

Proof. If $\overline{G}$ induces $K_1 \vee C_{2q+1}$, then G induces $K_1 \dot{\cup} \overline{C_{2q+1}}$. Lemmas 2.5 and 2.6 give eternal domination number four for that induced subgraph, contradicting Lemma 2.4. $\square$

Theorem 3.3 (template split). *Let G satisfy the hypothesis of Theorem 1.1, and put $H = \overline{G}$. Then H contains a hub-free induced C_ℓ for some*

$$\ell \in \{5, 7, 9\}. \tag{3.1}$$

Proof. Proposition 2.3 gives

$$\omega(H) = \alpha(G) = 3 < \chi(H) = \theta(G). \tag{3.2}$$

Thus H is imperfect, and the Strong Perfect Graph Theorem supplies an induced odd hole or odd antihole. If an odd antihole has length $2q + 1$, then $q = \omega(\overline{C_{2q+1}}) \leq 3$. It therefore has length five or seven. The five-antihole is C_5 . An induced $\overline{C_7}$ in H would induce C_7 in G , contradicting Lemmas 2.4 and 2.7. Hence H contains an odd hole.

The hole is hub-free by Proposition 3.2. Moreover, because $\gamma(G) = 3$, no pair dominates G . Equivalently, every pair of vertices of H has a common H -neighbor. The endpoints of a rim edge of an induced hole have no common neighbor on the rim, so every rim edge has a common neighbor outside the hole. If exactly one vertex were outside, that vertex would be adjacent to both endpoints of every rim edge, hence would be a hub. At least two vertices are therefore outside the hole. Since $|V(H)| = 12$, its odd length is at most ten, proving (3.1). $\square$

4 Exact finite encoding and coverage

We now describe the formulas used to exclude the three cases in Theorem 3.3. The description is part of the coverage proof: unsatisfiability of a formula is useful only after every graph in the claimed branch is shown to produce a satisfying assignment.

Fix $\ell \in \{5, 7, 9\}$. Label an induced cycle of $H = \overline{G}$ by $0, \dots, \ell - 1$ cyclically. The endpoints $0, 1$ have a common H -neighbor outside the rim; label one such vertex ℓ . The remaining vertices receive arbitrary labels. The template fixes the cycle, the edges $0\ell, 1\ell$, and a clause forbidding each outside vertex from being a hub.

All three formulas use the same 6,886 Boolean variables:

$$\begin{aligned}
 66 & : e_{uv} & (uv \in E(H)), \\
 660 & : w_{ab,c} & (c \text{ witnesses a common } H\text{-neighbor of } a, b), \\
 220 & : f_D & (D \in \binom{[12]}{3} \text{ is in the eternal family}), \\
 5,940 & : m_{D,r,u} & (u \in D \text{ responds to } r \notin D).
 \end{aligned} \tag{4.1}$$

The base clauses assert:

1. no four vertices form a clique in H ;
2. every vertex pair has a common H -neighbor, with each true witness implying its two incident H -edges;
3. the labeled induced, hub-free C_ℓ template;
4. every proper cut has a crossing G -edge, encoded as a negative H -edge;
5. every selected triple dominates G ;
6. at least one triple is selected;
7. for every selected D and every $r \notin D$, at least one $m_{D,r,u}$ is true, and every true move requires $ur \in E(G)$ and selects the successor $(D - \{u\}) \cup \{r\}$; and
8. every triangle of H is selected.

The last clauses are redundant but sound: a triangle of H is an independent triple of G , so it lies in every eternal three-family by Lemma 2.2. Notice also that a legal move is encoded by $\neg e_{ur}$, because the edge variables describe H , not G . The successor's family variable invokes the same domination clauses, so every resulting configuration dominates.

To encode $\chi(H) > 3$, let $c : [12] \rightarrow \{0, 1, 2\}$ be a coloring proper on the forced template edges, and append

$$C_c = \bigvee_{\substack{u < v \\ c(u)=c(v)}} e_{uv}. \tag{4.2}$$

This clause is false exactly when c is a proper coloring of H . Color names are canonicalized by first use. A complete bank for the ℓ -template has

$$\frac{(2^\ell - 2)3^{11-\ell}}{6} \tag{4.3}$$

rows: first color the odd rim, force the distinguished common neighbor to use the third color, color the remaining vertices freely, and divide by the free action of the six color permutations.

The C_5 formula has one further, sound symmetry breaker. For $v \in \{6, \dots, 11\}$, form the core signature

$$s(v) = (e_{0v}, e_{1v}, \dots, e_{5v}).$$

The complete formula is invariant under every permutation of these six vertices, provided graph, witness, family, and move variables are all relabelled. Hence every model has an orbit representative satisfying

$$s(6) \leq_{\text{lex}} s(7) \leq_{\text{lex}} \dots \leq_{\text{lex}} s(11). \quad (4.4)$$

Each adjacent comparison is encoded by 63 clauses, one for every possible first differing coordinate and common prefix, for 315 clauses in total. Thus (4.4) is an equisatisfiable orbit restriction, not a logical consequence silently added to the formula.

Lemma 4.1 (graph-to-formula coverage). *Suppose that G is a connected 12-vertex graph with*

$$\gamma(G) = \alpha(G) = \gamma^\infty(G) = 3 < \theta(G) \quad (4.5)$$

and that $\overline{G}$ contains a hub-free induced C_ℓ , where $\ell \in \{5, 7, 9\}$. Then the graph and an eternal three-family induce a satisfying assignment of the exact formula for that branch in Table 1.

Proof. Use the labeling above and set e_{uv} from the actual edges of $H = \overline{G}$. Equation (4.5) gives $\omega(H) = 3$, so the no- K_4 clauses hold. Since no pair dominates G , select an actual common H -neighbor for every pair and set the corresponding witness variable. Connectedness and hub-freeness satisfy the cut and template clauses.

Set f_D true exactly on an actual nonempty eternal three-family. For every selected D and $r \notin D$, select one guard promised by (2.2). This satisfies the domination, response, legal-edge, and successor clauses. Lemma 2.2 satisfies the triangle-selection clauses.

Finally, $\theta(G) = \chi(H) > 3$, so H fails every three-coloring. It therefore satisfies every appended clause (4.2), whether the branch uses a complete bank or only a recorded subset. For $\ell = 5$, permute the six undistinguished outer vertices, together with all auxiliary variables, to obtain an orbit-equivalent assignment satisfying (4.4). $\square$

Remark 4.2. The C_5 and C_7 formulas use complete coloring banks of respectively 3,645 and 1,701 clauses. The C_9 certificate arose after 170 coloring clauses in a counterexample-guided run. Completeness of that 170-row list is neither claimed nor needed: a target with $\chi(H) > 3$ satisfies every valid clause (4.2), including those 170.

5 Proof certificates and the finite theorem

A clause is a reverse-unit-propagation (RUP) consequence of a formula if unit propagation after adjoining the negation of that clause derives a contradiction. Appending a RUP consequence preserves satisfiability. Therefore an addition-only sequence of RUP consequences ending in the empty clause is a checkable unsatisfiability certificate [8].

Table 1 binds each formula and its accepted addition-only proof. Hashes are SHA-256. The C_5 proof is in binary DRAT format; the other two are in ASCII DRAT format. In every case the accepted replay used only RUP: the checker reported zero RAT lemmas.

The formula hashes, in C_5, C_7, C_9 order, are

```
c6a0811c718ff8e9352f253e4ce225ce2826def9c3e4a9cd55f0b0152703d104,
6a011e685e58ef517f2ab8253ca40987bd7b742a470bedbacdc3a5e94fc995a7,
2845f242a094484a8d114e70ca1a8678dfcfff79fadd56bd57813e25c2e49523d.
```

Table 1: Exact branch formulas and accepted addition-only RUP proofs.

	C_5	C_7	C_9
variables	6,886	6,886	6,886
clauses	23,968	21,718	20,200
literals	192,169	148,551	117,841
CNF bytes	754,323	621,864	530,053
proof additions	247,981	284,317	4,705
proof bytes	6,337,621	18,093,724	65,906

The corresponding proof hashes are

c6c24853e30073e66fb396441edb176a0160d062a8558e25fa18a955f33927c3,
e8052df40d3e0c39b945a8735889039daba55eacc351e1822828b3d94f7baae9,
24c5647d3a57f2de221fba96747c618575a3aba086c5e4bca17aade55ce7d4ab.

For C_5 , an independent parser verified that the retained proof is exactly the addition subsequence of the raw binary proof, with 245,439 deletions removed, and that its last addition is the empty clause. A fresh warning-fatal, forward, RUP-only replay used 10,912,555 resolution steps and ended in `s VERIFIED`. For C_7 , two strict warning-free RUP-only replays accepted the 284,317-addition proof. For C_9 , a separately written standard-library checker validated all 4,705 RUP additions and the final empty clause; a pinned external checker independently agreed. The exact paths, checker hash, acceptance records, and commands appear in Appendix A.

Proof of Theorem 1.1. Suppose G satisfies the displayed hypothesis. By Proposition 3.1, G is connected, and by Proposition 2.3 it satisfies (4.5). Theorem 3.3 gives a hub-free induced C_5 , C_7 , or C_9 in $\overline{G}$. Lemma 4.1 then gives a model of the exact corresponding formula in Table 1. The accepted RUP refutation proves that formula unsatisfiable, a contradiction. $\square$

Remark 5.1 (scope). The theorem certifies the complete $(n, k) = (12, 3)$ counterexample slice, including disconnected graphs. It does not certify the $n = 12, k \geq 4$ slices, any order $n \geq 13$, absence of counterexamples through order 12, or the universal conjecture.

6 Research and verification provenance

This manuscript and the associated research artifacts were produced with substantial assistance from autonomous AI systems. No mathematical claim is accepted on model output alone. The structural lemmas were subjected to separate adversarial mathematical reviews; the exact formulas were independently reconstructed; and the decisive proof streams were parsed and replayed by programs that do not import the search implementation’s transition or encoding core. Two independent reviews audited the final composition from the three branch certificates to Theorem 1.1. Human authors remain responsible for the submitted version and for inspecting the archived evidence.

Data availability

The archive accompanying this manuscript contains the formulas, proofs, standalone auditors, acceptance records, and exact hashes described below. A permanent public archive identifier is to be inserted before submission.

A Reproducibility and exact artifact bindings

All paths below are relative to the campaign archive root `gamma_theta_eternal_domination/`. The claim-level acceptance record is

```
results/order12_k3_exclusion_acceptance.json.
```

The theorem and its two independent coverage reviews are

```
math/lemmas/order12_k3_exclusion.md
reviews/order12_k3_exclusion_hostile_review.md
reviews/order12_k3_exclusion_second_review.md.
```

The exact formulas are:

```
results/synthesis_k3_hole5_signature_package/instance.cnf
results/synthesis_k3_template_bank_packages/hole7/instance.cnf
certificates/synthesis_k3_hole9_orphan_000170_recovery/
source/orphan-attempt-000170/instance.cnf.
```

The accepted proofs are:

```
results/synthesis_k3_hole5_signature_seed0_600s_binary/
proof.additions.bdrat
certificates/synthesis_k3_hole7_full_bank_seed0_addition_only_v2/
proof/addition-only.rup.drat
certificates/synthesis_k3_hole9_orphan_000170_recovery/
proof/addition-only.rup.drat.
```

The pinned `drat-trim` executable has SHA-256

```
31df522b8b2b71acd357723b0e826cf488826ed78ad9e3a7bcad241271812beb.
```

The C_5 source/run-gate revision is `6f3ef0a0970b7214c34018fe32ea1ceeb5764d17`; its 12 run artifacts were frozen without payload changes at `dff45f4239e4acabc461533a0a213beec18ec56d`. The complete C-035 acceptance package was committed at `36d8191a`.

From the campaign archive root, one fail-closed command performs the claim-level replay:

```
python3 -I -B repro/c035/replay.py --mode full \
--output /fresh/path/c035-full-replay.json
```

The wrapper first validates 91 exact file bindings, accepted Git objects, branch activations, theorem scope, and manifest rows. It then creates fresh private temporary directories and sequentially runs the two independent C_5 audits, the sealed C_7 proof replay, and the sealed C_9 proof replay. It accepts only exact warning-free status records and rehashes every input afterward. A successful result has status `PASS_FULL_C035_REPLAY`, `claim_status=CERTIFIED-FINITE`, and `proofs_freshly_replayed=true`. Time, memory, disk, and live-load gates reject rather than weaken a run. The fast mode validates metadata only and explicitly reports `NO_MATHEMATICAL_CLAIM`.

References

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